

SWARAJ PATIL

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Full-stack engineer and former technical lead with 5 years of experience across cloud (AWS, GCP), backend services, and React front-ends, now a graduate researcher at Northeastern (MScS, 3.84 GPA). Shipped production platforms end to end, led a product team, and built LLM- and RAG-powered tools. Currently writing a master's thesis on LLM-based data conversion, designing the AWS deployment for an open-source scientific platform used by labs worldwide, and building large-scale healthcare data pipelines in Python. Comfortable being dropped into an unfamiliar system and making it faster, clearer, and easier to run.

EDUCATION

Northeastern University, Khoury College of Computer Sciences

Jan 2026 - May 2028 (Expected)

M.S. in Computer and Information Science · Boston, MA

GPA: **3.84 / 4.0** · Coursework: Database Management Systems (A), Program Design Paradigm (A-), Algorithms, Cloud Computing, Master's Thesis

VES Institute of Technology

Aug 2017 - Jun 2021

Bachelor of Engineering · Mumbai, India · CGPA: 7.7 / 10

RESEARCH EXPERIENCE

Northeastern University, [Olga Vitek Lab](#), Khoury College

Jan 2026 - Present

Research Assistant (Part-Time)

Boston, MA

- **Master's thesis, MSstatsConvertLLM:** Building an LLM-based converter that infers a JSON schema mapping from a tool's column headers and sample rows, replacing hand-written per-format converters. Designed the mapping-correctness metric scoring inferred mappings against deterministic converters as ground truth, benchmarked across ~7 input formats and multiple models. First-author technical note targeting the *Journal of Proteome Research*.
- **Cloud scalability:** Designed the AWS deployment for [MSstatsShiny](#), an open-source R Shiny analysis platform used by the global research community: ECS Fargate behind an Application Load Balancer, Python CDK for infrastructure as code, and CodePipeline CI/CD, sized for 50 concurrent workshop users after auditing the single-instance production setup.
- **Feature development:** Shipped features end to end across a family of open-source R packages over 8 months, from spec through peer code review to merge: a simulation-based sample-size calculator, an interactive analysis module that lets scientists run advanced statistics without writing code, an upload path so users can bring already-processed data straight into analysis, and an upload page that reads a file's first rows to populate its form fields automatically.
- **New domain support:** Led the work extending the platform beyond its original protein-only scope to a second class of scientific data (metabolomics) across five merged pull requests: a new file-format upload and conversion path, results tables and network-visualization views that adapt to the new data type, and a rewrite of 23 interface strings so the product's language generalizes across domains.
- **Open-source maintenance:** Wrote new data-format converters (wide-to-long reshaping, null-safe aggregation) and unblocked every fresh install of the toolchain by replacing a logging dependency that had been pulled from its package repository, reimplementing it on the standard library while preserving byte-identical output; also refactored an ~800-line module into focused helpers and fixed a cross-cutting bug that had broken data previews for all workflows.
- Co-coordinate the lab's annual May Institute training program, an international computational-analysis course for practicing scientists.

Ramped up on proteomics and mass spectrometry from scratch in one semester to ship research-grade software, and ran a live platform demo for an international short-course collaborator.

Northeastern University, [Supply Chain & Information Management Group](#), D'Amore-McKim School

Jun 2026 - Aug 2026

Research Assistant (Part-Time)

Boston, MA

- Built the full empirical pipeline for three healthcare-quality studies on CMS Medicare data (skilled-nursing facilities and home-health agencies) in Python, pandas, and statsmodels, from raw federal files to journal-ready regression tables.
- Engineered reproducible provider panels by joining provider, cost-report, utilization, quality, and market-saturation sources on provider ID and year, producing a 47,000-row, 90-plus-column analysis dataset.
- Delivered 15 publication-format tables with clustered standard errors and state and year fixed effects; showed the chain-affiliation quality penalty is fully mediated by nurse staffing, with RN hours the dominant driver.
- Designed a difference-in-differences test isolating genuine post-citation improvement from reporting lag and regression to the mean, finding ~90% of the raw improvement was regression to the mean; rebuilt a mis-specified market-competition measure from a ZIP-to-CBSA crosswalk after auditing the source variable.

PROFESSIONAL EXPERIENCE

KPMG (Client: ZS Associates / Amgen)

Sep 2025 - Dec 2025

Consultant, Full-Stack Developer

Pune, India

- Led the front-end build of the Work Centre Team dashboard on Amgen's ASM platform (React, FastAPI, PostgreSQL), visualizing SQDIP metrics (Safety, Quality, Delivery, Inventory, Productivity) for operational decision-makers.
- Designed a three-tier hierarchical drill-down filter enabling site-level, pillar-wise KPI exploration.

Beellogical Software Solutions

Aug 2023 - Aug 2025

Technical Lead

Pune, India

- Delivered IncBuddy, a React, NestJS, and PostgreSQL travel platform: built XLSX-driven bulk admin uploads, dynamic PDF generation, Google Maps integration, and Razorpay payments; added Redis caching and owned deployment on AWS.
- Led front-end engineering on IRYS Cloud, a Next.js insurance platform running on GCP: published a custom UI and icon library, built dynamic validated forms, and shipped utilities like image cropping and address validation.
- Steered architecture, hiring, and client communication as technical lead to ship scalable web solutions on schedule.

Capgemini Technology Services India Ltd.

Jul 2021 - Aug 2023

Senior Software Engineer

Pune, India

- Built React UIs for Fiserv banking platforms using custom hooks, higher-order components, and the Context API; implemented dynamic data filtering and API integrations across production financial systems.

SELECTED PROJECTS

PolicyLens · RAG Platform for University Governance Documents · ChromaDB, Ollama/Groq, Docker [Live](#) · [Demo](#) · [GitHub](#)

- Built a retrieval-augmented Q&A system over university faculty-handbook PDFs (Northeastern, BU, Harvard) that answers plain-language policy questions and runs side-by-side cross-institution comparisons.
- Enforced page-anchored citations at both the prompt and parser layers so every answer traces to its source page, and flagged passage-level reading difficulty (Flesch-Kincaid) for accessibility.
- Built a pluggable LLM provider layer (Claude, Gemini, Ollama, Groq) and deployed the full stack on Vercel and Render.

Retrace · RAG-Native LLM Observability Platform (in progress) · Python, FastAPI, ClickHouse, PostgreSQL, Next.js [GitHub](#)

- Building an open-source, self-hostable observability platform for LLM apps that treats retrieval as a first-class trace primitive (chunk-level scores, embedding model, retrieval latency, citation groundedness), where tools like Langfuse and Phoenix treat retrieval as a black box.
- Designing OpenTelemetry-compatible ingestion with auto-instrumentation for the OpenAI and Anthropic SDKs, backed by ClickHouse for high-cardinality trace data and PostgreSQL for metadata, with Docker Compose for one-command self-hosting.

TrialCompanion AI · Clinical Trial Intelligence Platform · React, FastAPI, PostgreSQL, Gemini [Demo](#) · [GitHub](#)

- Built an end-to-end platform that turns complex clinical-trial protocols into patient-friendly insights using LLM-powered parsing and summarization.
- Engineered a protocol ingestion pipeline that extracts eligibility criteria, schedules, and medical entities from PDFs, and a context-grounded Q&A chatbot constrained to protocol content to prevent hallucination.

Root-Cause Analysis of a MongoDB TTL Performance Bug · Independent Research, Northeastern [Report](#)

- Reproduced and root-caused a 3.5x to 6.6x regression in MongoDB TTL deletions, tracing execution across the TTL monitor, query planner, and storage engine.
- Built cursor-level instrumentation, identified inefficient backward index traversal as the cause, and proposed a forward-scan fix plus per-collection observability metrics; correlated findings with OSDI '23 research.

TECHNICAL SKILLS

Cloud & DevOps: AWS (ECS Fargate, ALB, CDK, CodePipeline), GCP, Docker, CI/CD, Vercel

Backend: Python (FastAPI, pandas, statsmodels), Node.js, Express.js, NestJS, REST, GraphQL

AI / ML & Data: LLM integration (Claude, Gemini, Groq, Ollama), retrieval-augmented generation (RAG), prompt engineering and evaluation, ChromaDB, R (Shiny, Bioconductor)

Front-End: React.js, Next.js, TypeScript, Redux, Tailwind, Material UI

Databases: PostgreSQL, MongoDB, MySQL, Redis, ClickHouse, Supabase

Languages: Python, JavaScript, TypeScript, R, Java, C, SQL

Tools: Git, JIRA, Swagger/OpenAPI, Figma